Statistical Methods in AI

Assignment 3: Build a Real ML App

13 Themes · 103 Variants · Pick One

IIIT Hyderabad · Academic Year 2025–26

Read this catalogue, discuss with your team, and submit your top 3 preferences via the assignment form.

Time budget: 4–6 hours per person (total).

Compute: Google Colab free tier (T4 GPU) or any laptop. No paid GPUs needed.

Deliverables: (i) a working web app (Streamlit / Gradio / HF Space) (ii) a 4–6 page report (iii) GitHub repo.

Tier 1: 1–3 students. Smaller dataset, simpler scope, lighter report.

Tier 2: 5–7 students. Larger dataset, deeper evaluation, ICVGIP-style report.

Use of LLMs (ChatGPT, Claude, Gemini) for data labelling, code scaffolding, and report drafting is encouraged.
You must disclose what you used and explain every line of your own code in the viva.

Contents

Before You Pick

Why this assignment exists

Every topic here:

- Solves a problem a non-CS person would understand.
- Has a clear, accessible dataset (link given for every variant).
- Ends in a demo you can share on LinkedIn.
- Fits in the time budget without burning your week.

What you will hand in

1. **GitHub repository.** Public, with `README.md`, `requirements.txt`, training notebook, app code.
2. **Live demo.** Deployed on HuggingFace Spaces (free) or Streamlit Community Cloud (free). If hosting fails, a recorded ≤2-minute screen demo is acceptable.
3. **Technical report (PDF).** Sections: Introduction, Data, Method, Results, Limitations, References. Include an Ablation section where necessary.
4. **One-slide pitch.** Single PNG/PDF you could post on LinkedIn.

The two tiers

	Tier 1 (easier)	**Tier 2 (more involved)**
Team size	1–3 students	5–7 students
Per-person time	4–6 hours	4–6 hours
Dataset scale	200–2,000 examples	2,000–20,000 examples
Models	Pre-trained / zero-shot / 1-epoch fine-tune	Light fine-tune + ablation + comparison
Report	4–6 pages, lab-report style	6–8 pages, ICVGIP-style
App polish	Working Streamlit demo	Polished + deployed on HF Spaces
Maximum marks	100	100

Tier 2 is not "worth more marks" — the ceiling is the same. Tier 2 simply asks more from a larger team. It may be more fun and engaging and would generally result in deeper insights and learnings.

Using LLMs (encouraged!)

You may freely use ChatGPT / Claude / Gemini / Copilot for: data labelling (paste 50 reviews, ask for sentiment labels), synthetic data generation, code scaffolding (Streamlit UI, dataset loaders), debugging, and report drafting. **Honesty rules:** (a) cite the LLMs you used in Acknowledgements, (b) all evaluation and analysis must be your own.

How to choose and submit

1. Read the 13 themes. Skim variants in the ones that interest you.
2. Form your team (pick a tier first — it sets the size).

3. Submit via the Google Sheet (https://docs.google.com/spreadsheets/d/1E6P6lo_E4S-YG61YbZmCb-slnJa5bUCRkLq4w/edit?usp=sharing). Only select from the available ones which have not already been taken (from the google sheet) - fist come first serve basis.
4. **Want to propose your own project?** The form has an optional section. Faculty/TA approval required.

The 13 Themes

#	Theme	Type	Variants
T1	Crop Disease Detection	Image cls	8
T2	Indian Food Recognizer	Image cls	8
T3	Handwritten Indic Script Recognition	Image $\rightarrow$ text	8
T5	Road Safety & Traffic Vision	Detection	8
T6	Yoga & Fitness Pose Coach	Pose + cls	8
T7	Indian Wildlife & Bird Identifier	Image cls	8
T8	Indic Review & Sentiment Analyzer	NLP	8
T9	News Classifier & Summarizer	NLP + LLM API	8
T10	Government-Services RAG Chatbot	RAG + LLM API	8
T11	Indic Speech Command Recognizer	Audio cls	7
T12	Monuments & Heritage Identifier	Image cls	8
T13	Smart Document Parser	LLM API	8
T14	Personalised Indian Recommender	Embeddings + LLM	7
	Total		**103**

Each theme is described once; the variants are listed at the end of the theme as a table. Pick one.

T1 — Crop Disease Detection

The problem. A farmer photographs a sick leaf with their phone. Your app names the disease and suggests a next step. Indian agriculture loses crores every year to misdiagnosed crop disease.

What you will build. Streamlit / Gradio web app: upload leaf photo → disease prediction → confidence bar → farmer-friendly explanation paragraph and recommended action.

Skills. Image classification, transfer learning, data augmentation, simple web UIs.

Common base dataset (used by most variants). PlantVillage — 38 classes, 54,303 images of healthy and diseased leaves across 14 crops (Mohanty et al. 2016). Available at:

- Kaggle: `kaggle.com/datasets/abdallahalidev/plantvillage-dataset`
- HuggingFace: `hf.co/datasets/mohanty/PlantVillage`

Approach. Filter PlantVillage to the relevant crop subset, fine-tune a small `timm` model (`efficientnet_b0` or `mobilenetv3_small`) for 3 epochs. For disease descriptions, ask Gemini once for a 2-sentence farmer-friendly description per class and cache as JSON.

Variants

ID	Variant	Tier	Specific dataset
T1.1	Tomato disease	**Tier 1**	PlantVillage tomato subset (10 classes), within `kaggle.com/datasets/abdallahalidev/plantvillage-datas`
T1.2	Rice disease	**Tier 1**	`kaggle.com/datasets/vbookshelf/rice-leaf-diseases` (4 classes: bacterial blight, blast, brown spot, tungro).
T1.3	Potato disease	**Tier 1**	PlantVillage potato subset (3 classes: early blight, late blight, healthy).
T1.4	Cotton disease	**Tier 1**	`kaggle.com/datasets/janmejaybhoi/cotton-disease-datas` (4 classes: diseased/fresh leaf and plant).
T1.5	Sugarcane disease	**Tier 1**	`kaggle.com/datasets/nirmalsankalana/sugarcane-leaf-di` (multiple disease classes).
T1.6	Pepper / chili disease	**Tier 1**	PlantVillage pepper subset (bacterial spot + healthy).
T1.7	Multi-crop unified model	**Tier 2**	Full PlantVillage (38 classes); routing model: predict crop → predict disease.
T1.8	Mobile / PWA version	**Tier 2**	Any of the above; export MobileNetV3 to TFLite or build as a PWA.

T2 — Indian Food Recognizer

The problem. Point your phone at a thali; the app names every dish, gives a calorie estimate, and flags allergens (nuts, dairy, gluten).

What you will build. Streamlit app: upload food photo → predicted dish → approximate calories → colour-coded allergen badges.

Skills. Image classification, transfer learning, joining image labels to a metadata table.

Common base datasets.

- *Master image dataset:* kaggle.com/datasets/iamsouravbanerjee/indian-food-images-dataset (4,000 images, 80 classes). All image-classification variants subset this.
- *Metadata (region, course, diet, ingredients) for 255 dishes:* kaggle.com/datasets/nehaprabhavalkar/ind Use as the join table for nutrition / diet info.
- *Recipes corpus:* kaggle.com/datasets/kanishk307/6000-indian-food-recipes-dataset (with cuisine, course, diet, ingredients).
- *Smaller alternative:* kaggle.com/datasets/theeyeschico/indian-food-classification (top 20 dishes).

For nutrition data, the Indian Food Composition Tables (IFCT 2017) PDF is available from the National Institute of Nutrition (https://www.nin.res.in); or generate once via Gemini and cache as JSON.

Variants

ID	Variant	Tier	Specific dataset
T2.1	North Indian curries	**Tier 1**	15 curry classes from kaggle.com/datasets/iamsouravbanerjee/indian-food-imag
T2.2	South Indian breakfast	**Tier 1**	10 classes (idli, dosa, vada, upma, pongal, etc.) from the same master set.
T2.3	Indian sweets	**Tier 1**	15 mithai classes from the master set.
T2.4	Street food	**Tier 1**	12 classes (pani puri, vada pav, chaat, etc.) from the master set.
T2.5	Indian breads	**Tier 1**	8 classes (roti, naan, paratha, kulcha, bhatura, etc.) from the master set.
T2.6	Regional thali identifier	**Tier 1**	Master set, regrouped by region using the metadata in kaggle.com/datasets/nehaprabhavalkar/indian-food-101.
T2.7	80-class + nutrition app	**Tier 2**	Full kaggle.com/datasets/iamsouravbanerjee/indian-food- + nutrition join.
T2.8	Diabetic meal scorer	**Tier 2**	Master set + IFCT nutrition data; rule-based scoring on classifier output.

T3 — Handwritten Indic Script Recognition

The problem. Write a character on the screen (or upload an image) and get the Unicode prediction. Useful for educational apps, form digitization, and accessibility.

What you will build. Streamlit app with a drawable canvas (`streamlit-drawable-canvas`): draw a character with your mouse → live prediction. Optional "practice mode" shows a target character and gives feedback.

Skills. Small CNNs, training from scratch, Streamlit canvas component.

Approach. A 3-layer CNN (∼300k params) trained from scratch reaches 97%+ in <5 minutes on Colab T4. No transfer learning needed.

Variants

ID	Variant	Tier	Specific dataset
T3.1	Devanagari digits	**Tier 1**	UCI Devanagari Handwritten Character Dataset (id 389), digits subset (10 classes, 20,000 images).
T3.2	Devanagari characters	**Tier 1**	Same UCI dataset, full set: 46 classes, 92,000 images.
T3.3	Tamil characters	**Tier 1**	`kaggle.com/datasets/faizalhajamohideen/uthcdtamil-h` (uTHCD, 156 character classes).
T3.4	Telugu vowels	**Tier 1**	`kaggle.com/datasets/syamkakarla/telugu-6-vowel-data` (6 vowel classes; small but complete).
T3.5	Bengali characters	**Tier 1**	BanglaLekha-Isolated (84 classes, 166k images): data.mendeley.com/datasets/hf6sf8zrkc/2.
T3.6	Kannada-MNIST digits	**Tier 1**	kaggle.com/competitions/Kannada-MNIST (10 classes, 70k images).
T3.7	Multi-script router	**Tier 2**	Combine UCI Devanagari, uTHCD Tamil, BanglaLekha and Kannada-MNIST. First model predicts script, second predicts character.
T3.8	Printed-text textbook OCR	**Tier 2**	Use Tesseract (`tesseract-ocr`) with Indic language packs + a fine-tuned image preprocessor. Build a "scan a school textbook page" app.

T5 — Road Safety & Traffic Vision

The problem. Indian roads have unique hazards. Build apps that detect traffic signs, helmet-less riders, potholes, vehicles, etc. from a photo or short video.

What you will build. Streamlit app: upload image or short video → annotated output with bounding boxes → count summary.

Skills. Object detection (YOLOv8), pre-trained model usage, video frame processing.

Common base.

- `ultralytics YOLOv8n` (3M params) is pre-trained on COCO and already detects vehicles and persons.
- Roboflow Universe (`https://universe.roboflow.com`): hundreds of free YOLO-format datasets. Search "India helmet", "India pothole", etc.

Variants

ID	Variant	Tier	Specific dataset
T5.1	Indian traffic sign classifier	**Tier 1**	`kaggle.com/datasets/sarangdilipjodh/indian-traffi` (85 classes) or `kaggle.com/datasets/neelpratiksha/indian-traffic-`
T5.2	Helmet detector	**Tier 1**	Roboflow Universe "helmet detection" (search at `https://universe.roboflow.com/search?q=helmet`).
T5.3	Pothole spotter	**Tier 1**	Roboflow Universe "pothole detection" (`https://universe.roboflow.com/search?q=pothole`).
T5.4	Vehicle counter	**Tier 1**	No new dataset — pre-trained YOLOv8n on COCO already detects car/bus/truck/motorbike. Test on any traffic video.
T5.5	Lane-line detector	**Tier 1**	No labelled dataset needed — pure OpenCV (Hough transform). Test on dash-cam clips from YouTube.
T5.6	License plate cropper + OCR	**Tier 2**	Roboflow Universe "Indian license plate" YOLO dataset + PaddleOCR / EasyOCR for text.
T5.7	Road safety dashboard	**Tier 2**	Combine 3+ of the above datasets; per-image safety score.
T5.8	Dashcam highlight extractor	**Tier 2**	Use COCO-pretrained YOLOv8n + simple heuristics on detection-box trajectories from a 1-min clip.

T6 — Yoga & Fitness Pose Coach

The problem. A webcam-based pose coach: tells the user which yoga asana they are attempting and whether their form is correct.

What you will build. Streamlit app with `streamlit-webrtc` for live webcam (or upload an image). Show: predicted asana, keypoint overlay, form feedback.

Skills. MediaPipe pose estimation (no training needed), simple MLP classifier on keypoints, rule-based form checks.

Common base.

- Pose extraction: `mediapipe.solutions.pose` (`pip install mediapipe`). 33 keypoints, real-time on CPU.
- For form-correctness rules, ask Gemini: *"For Vrikshasana, give 5 measurable form rules based on body keypoints."*

Variants

ID	Variant	Tier	Specific dataset
T6.1	5-asana classifier	**Tier 1**	`kaggle.com/datasets/niharika41298/yoga-poses-da` (5 classes).
T6.2	Surya Namaskar step counter	**Tier 1**	12-class subset of `kaggle.com/datasets/shrutisaxena/yoga-pose-imag` (47 asanas total — pick the 12 SN poses).
T6.3	Pushup rep counter	**Tier 1**	No labelled data needed — use MediaPipe elbow-angle on any pushup video. Test on YouTube clips.
T6.4	Squat form checker	**Tier 1**	No labelled data needed — MediaPipe knee/hip angles on squat videos.
T6.5	Plank timer + form check	**Tier 1**	No labelled data needed — MediaPipe hip-line straightness on plank videos.
T6.6	20-asana classifier + form coach	**Tier 2**	20-class subset of `kaggle.com/datasets/shrutisaxena/yoga-pose-imag`
T6.7	Personalised home-yoga app	**Tier 2**	Same 47-class dataset; user picks goals, app builds routine, monitors form.
T6.8	Desk-posture monitor	**Tier 2**	No labelled data needed — continuous MediaPipe; alert if user slouches >30s.

T7 — Indian Wildlife & Bird Identifier

The problem. Snap a photo on a forest trek → get the species name, conservation status, and a fun fact.

What you will build. Streamlit app: upload → top-3 species predictions → Wikipedia paragraph → IUCN status badge.

Skills. CLIP zero-shot classification (no training!), or light fine-tuning.

Common base. For zero-shot, use `openai/clip-vit-base-patch32`. For light fine-tuning, use `efficientnet_b0` from `timm`. Fun facts + IUCN status: ask Gemini once, cache as JSON.

Variants

ID	Variant	Tier	Specific dataset
T7.1	Indian birds (25 species)	**Tier 1**	`kaggle.com/datasets/arjunbasandrai/25-indian-bir` (37k images, 25 species).
T7.2	Indian mammals	**Tier 1**	`kaggle.com/datasets/asaniczka/mammals-image-clas` (45 mammals; filter to species native to India).
T7.3	Snakes (venomous vs not)	**Tier 1**	`kaggle.com/datasets/adityasharma01/snake-dataset` (binary venomous/non-venomous, Indian species). *Add a clear medical disclaimer.*
T7.4	Butterflies	**Tier 1**	`kaggle.com/datasets/phucthaiv02/butterfly-image-` (75 species) or `kaggle.com/datasets/gpiosenka/butterfly-images40` (100 species).
T7.5	Big cats (4–5 species)	**Tier 1**	`kaggle.com/datasets/anshulmehtakaggl/wildlife-an` — filter to tiger / leopard / lion / clouded leopard. CLIP zero-shot also works.
T7.6	90-animal general identifier	**Tier 1**	`kaggle.com/datasets/iamsouravbanerjee/animal-ima` (90 species, sufficient as-is).
T7.7	100-species fine-tuned identifier	**Tier 2**	Combine the 45-mammal + 25-bird + 30-other datasets above into a single ∼100-class model.
T7.8	Camera-trap auto-tagger	**Tier 2**	Same combined dataset; build a batch-folder interface for forest researchers (folder → CSV).

T8 — Indic Review & Sentiment Analyzer

The problem. Indians review products and services in Hindi, Tamil, Hinglish, etc. Build a sentiment analyzer for one language and surface insights for a small business owner.

What you will build. Streamlit app: paste a single review (label + confidence), *or* upload a CSV (label all, dashboard with pie chart and top themes).

Skills. Multilingual NLP, fine-tuning small BERT-family models, or zero-shot classification with hosted LLM APIs.

Common base.

- AI4Bharat IndicSentiment (13 Indic languages, parallel reviews): `hf.co/datasets/ai4bharat/IndicSenti`
- Models: `ai4bharat/indic-bert` (110M) for fine-tuning; or zero-shot via free Gemini / Groq Llama-3 API.

Variants

ID	Variant	Tier	Specific dataset
T8.1	Hindi product review	**Tier 1**	`hf.co/datasets/ai4bharat/IndicSentiment` (Hindi config).
T8.2	Tamil restaurant review	**Tier 1**	`hf.co/datasets/ai4bharat/IndicSentiment` (Tamil config).
T8.3	Telugu movie review	**Tier 1**	`hf.co/datasets/ai4bharat/IndicSentiment` (Telugu config).
T8.4	Bengali book review	**Tier 1**	`hf.co/datasets/ai4bharat/IndicSentiment` (Bengali config).
T8.5	Hinglish e-commerce	**Tier 1**	L3Cube HingCorpus on GitHub: `https://github.com/l3cube-pune/code-mixed-nlp` + LLM-generated labels (Gemini for 500 examples).
T8.6	Marathi tweet sentiment	**Tier 1**	`hf.co/datasets/ai4bharat/IndicSentiment` (Marathi config); model: `l3cube-pune/marathi-bert-v2`.
T8.7	Multi-language analyzer + dashboard	**Tier 2**	Use 5+ language configs of `hf.co/datasets/ai4bharat/IndicSentiment`; theme extraction via LLM API.
T8.8	Aspect-based sentiment	**Tier 2**	`hf.co/datasets/ai4bharat/IndicSentiment` (any language; uses ASPECTS field) + LLM API for aspect extraction.

T9 — News Classifier & Summarizer

The problem. A daily-briefing app: pulls Indian news from RSS, classifies into categories, generates a 3-line summary per article.

What you will build. Streamlit app that on each load fetches the latest 30 articles, runs a zero-shot classifier and an LLM summary, shows category tabs (Politics / Sports / Tech / Business / Entertainment).

Skills. RSS parsing, zero-shot NLP, LLM API usage, simple caching.

Common base.

- Live RSS feeds (no scraping needed): The Hindu, NDTV, Indian Express. `pip install feedparser`.
- Static labelled corpus for training/eval: `kaggle.com/datasets/therohk/india-headlines-news-dataset` (21 years of Indian headlines).
- Models: `facebook/bart-large-mnli` (zero-shot classifier) + free Gemini 1.5 Flash API for summary.

Variants

ID	Variant	Tier	Specific dataset
T9.1	English news daily briefing	**Tier 1**	Live RSS + `kaggle.com/datasets/therohk/india-headlines-ne` for training/eval.
T9.2	Hindi news daily briefing	**Tier 1**	L3Cube-IndicNews Hindi config: `https://github.com/l3cube-pune/indic-nlp`.
T9.3	Tech news tracker	**Tier 1**	Live RSS filtered to tech outlets (TechCrunch, The Verge, YourStory). Eval set: filter `kaggle.com/datasets/therohk/india-headlines-ne`
T9.4	Sports highlights briefing	**Tier 1**	ESPN Cricinfo / Cricbuzz RSS. Eval: filter `kaggle.com/datasets/therohk/india-headlines-ne`
T9.5	Stock-market news → ticker	**Tier 1**	Live RSS from MoneyControl + Economic Times. Ticker list from NSE: `https://www.nseindia.com/market-data/securities-available-for-trading`.
T9.6	Translated regional news	**Tier 1**	Live English RSS + free Gemini API for translation; eval against L3Cube-IndicNews.
T9.7	Bias spectrum visualiser	**Tier 2**	Live RSS from 5 outlets; same-story matching via sentence-transformers embeddings.
T9.8	Personalised newsletter generator	**Tier 2**	Live RSS + L3Cube-IndicNews categories + LLM-generated email-style digest.

T10 — Government-Services RAG Chatbot

The problem. Indian government services (passport, voter ID, RTI, taxes, etc.) have lots of public PDF documentation. Build a chatbot that answers "how do I apply for X" by retrieving from official PDFs.

Why this is the easiest theme. No model training at all. You build a RAG (Retrieval-Augmented Generation) pipeline using free embeddings + a free LLM API.

What you will build. Streamlit chat (`st.chat_input`). Each answer shows source PDF citations. Optional: "Talk in Hindi" toggle.

Skills. Document parsing, embeddings, vector search, prompt engineering, chat UI.

Common base.

- Embeddings: `sentence-transformers/all-MiniLM-L6-v2` (free, CPU).
- Vector store: `chromadb` or `faiss-cpu`.
- LLM: free Google AI Studio (Gemini 1.5 Flash) or Groq (Llama-3 free).

Variants

ID	Variant	Tier	Source PDFs (the "dataset" is the PDF set you download once)
T10.1	Passport assistant	**Tier 1**	`https://www.passportindia.gov.in` (download 5–10 "Instruction Booklet" / FAQ PDFs).
T10.2	Voter ID / EPIC assistant	**Tier 1**	Election Commission of India: `https://www.eci.gov.in` (Form-6/7/8 instruction PDFs).
T10.3	RTI filing helper	**Tier 1**	`https://rti.gov.in` (RTI Act PDF + state-rules PDFs).
T10.4	Income tax FAQ chatbot	**Tier 1**	`https://www.incometax.gov.in` (ITR forms + instruction booklets).
T10.5	Driving licence & RC	**Tier 1**	`https://parivahan.gov.in` (Sarathi/Vahan PDFs).
T10.6	Ayushman Bharat eligibility	**Tier 1**	National Health Authority: `https://nha.gov.in` (PM-JAY guidelines PDFs).
T10.7	Multi-service unified + Hindi support	**Tier 2**	5+ of the above sources combined; language toggle, conversation memory.
T10.8	IIIT-H rules / hostel / academic bot	**Tier 2**	Public IIIT-H academic regulations + hostel rules PDFs from `https://www.iiit.ac.in`.

T11 — Indic Speech Command Recognizer

The problem. A small set of voice commands in an Indian language — useful for voice-controlled apps, accessibility tools, smart-home prototypes.

What you will build. Streamlit app with hold-to-record button: release → predicts command → triggers an action.

Skills. Audio processing, ASR, simple post-processing.

Common base.

- AI4Bharat IndicVoices (22 languages, 11,200+ hours transcribed): hf.co/datasets/ai4bharat/IndicVoic
- Google FLEURS (Indian splits: `hi_in`, `ta_in`, `te_in`, `bn_in`, `mr_in`, `gu_in`, `kn_in`, `ml_in`, `pa_in`): hf.co/datasets/google/fleurs.
- Self-recording: each team member records each command ∼10× on their phone. 3 students × 10 commands × 10 takes = 300 clips, plenty for fine-tuning.

Note. Mozilla Common Voice is no longer hosted on HuggingFace (moved to Mozilla Data Collective in October 2025). Use the AI4Bharat / FLEURS sources above.

Easiest path. Whisper-tiny (39M, runs on CPU) for transcription + string match. *Zero training.*

Variants

ID	Variant	Tier	Specific dataset
T11.1	Hindi digits (0–9)	**Tier 1**	hf.co/datasets/google/fleurs `hi_in` config + self-recorded digit utterances.
T11.2	Tamil yes/no/cancel/back/help	**Tier 1**	hf.co/datasets/google/fleurs `ta_in` for pre-training; self-recorded commands for the 5-class task.
T11.3	Hindi smart-home commands	**Tier 1**	hf.co/datasets/ai4bharat/IndicVoices Hindi config + self-recorded 8 commands.
T11.4	Bengali navigation commands	**Tier 1**	hf.co/datasets/google/fleurs `bn_in` + self-recorded 5 commands.
T11.5	Telugu emergency phrases	**Tier 1**	hf.co/datasets/google/fleurs `te_in` + self-recorded “call doctor / help / police”. Accessibility framing.
T11.6	Marathi kiosk navigation	**Tier 1**	hf.co/datasets/google/fleurs `mr_in` + self-recorded kiosk commands.
T11.7	Wakeword + multi-language commands	**Tier 2**	Whisper-tiny on combined FLEURS Indic configs + self-recorded “Hey Bharat” wakeword.

T12 — Monuments & Heritage Identifier

The problem. A tourist points their phone at a monument; the app names it, gives history, opening hours, and ticket prices.

What you will build. Streamlit app: upload photo → predicted monument → history paragraph → visit info card → "open in Google Maps".

Skills. CLIP zero-shot or light fine-tuning, Wikipedia scraping for metadata.

Common base.

- Main labelled dataset: `kaggle.com/datasets/danushkumarv/indian-monuments-image-dataset` (24 monuments, ~3.5k images).
- For variants beyond these 24, use Wikimedia Commons category pages (e.g. `https://commons.wikimedia.org/wiki/Category:Forts_in_Maharashtra`) — image URLs are direct download.
- For zero-shot: `openai/clip-vit-base-patch32`; CLIP needs no training data, just a class-name list.
- Metadata (history, hours, tickets): scrape Wikipedia or ask Gemini once and cache as JSON.

Variants

ID	Variant	Tier	Specific dataset
T12.1	Top monuments fine-tuned	**Tier 1**	`kaggle.com/datasets/danushkumarv/indian` (24 classes, ready to train).
T12.2	Forts of Maharashtra	**Tier 1**	CLIP zero-shot on 12-fort class list (Raigad, Sinhagad, Pratapgad, etc.); demo images from Wikimedia Commons — Forts in Maharashtra.
T12.3	Temples of Tamil Nadu	**Tier 1**	CLIP zero-shot on 15 famous TN temples; images from Wikimedia Commons — Hindu temples in Tamil Nadu.
T12.4	Mughal architecture identifier	**Tier 1**	CLIP zero-shot on 15 Mughal monuments; images from Wikimedia Commons — Mughal architecture in India.
T12.5	Hampi monuments	**Tier 1**	CLIP zero-shot on ∼10 Hampi structures; images from Wikimedia Commons — Group of Monuments at Hampi.
T12.6	Stepwells of India	**Tier 1**	CLIP zero-shot on ∼8 stepwells (Rani-ki-Vav, Chand Baori, Adalaj, etc.); Wikimedia Commons — Stepwells in India.
T12.7	100-monument fine-tuned + audio guide	**Tier 2**	`kaggle.com/datasets/danushkumarv/indian` + augment with Wikimedia Commons categories; LLM-generated audio narration via `gTTS`.
T12.8	Monument scavenger hunt game	**Tier 2**	Same combined source; gamified — app gives clues, user photographs the right monument.

T13 — Smart Document Parser

The problem. A user uploads a resume / invoice / certificate; the app extracts a clean structured JSON. Used heavily by HR-tech, fintech, and edtech startups in India.

What you will build. Streamlit app: upload PDF/image → structured form (editable) → download JSON/CSV. Tier 2: batch mode for bulk processing.

Skills. OCR, LLM prompting with JSON schemas, schema validation (Pydantic).

Common base.

- OCR: `pypdf` for text PDFs, `paddleocr` or `easyocr` for images.
- LLM (no training): free Gemini or Groq Llama-3 API with a strict JSON schema in the prompt.

Variants

ID	Variant	Tier	Specific dataset
T13.1	Resume parser	**Tier 1**	`kaggle.com/datasets/snehaanbhawal/resume-dataset` (2,400+ resumes with category labels) or `kaggle.com/datasets/dataturks/resume-entities-fo` (NER-annotated).
T13.2	GST invoice parser	**Tier 1**	SROIE: `kaggle.com/datasets/urbikn/sroie-datasetv` (1,000 receipt images with OCR + key-info JSON ground truth).
T13.3	Marksheet parser	**Tier 1**	No public dataset (privacy). Use 5–10 self-photographed CBSE/ICSE specimen marksheets, plus ask Gemini to generate 50 synthetic ones from a template.
T13.4	Lease / rent agreement parser	**Tier 1**	Generate 30 synthetic leases from a Word template + LLM-filled fields (Gemini prompt: *"Generate 30 fake but realistic 11-month residential lease agreements for Hyderabad with varying parties, rent amounts, terms."*).
T13.5	Prescription parser	**Tier 1**	`kaggle.com/datasets/mehaksingal/illegible-medica` or `kaggle.com/datasets/mamun1113/doctors-handwrit` Include disclaimer.
T13.6	Bank statement parser	**Tier 1**	No public dataset (privacy). Use 5–10 self-redacted statements + 50 synthetic ones (Gemini prompt: *"Generate 50 synthetic Indian bank statement transaction lists with date / description / debit / credit / balance."*).
T13.7	HR-tech mini suite	**Tier 2**	Combine `kaggle.com/datasets/snehaanbhawal/resume` + synthetic offer letters + synthetic payslips into one app.
T13.8	Bulk doc-to-database	**Tier 2**	Use SROIE `kaggle.com/datasets/urbikn/sroie-datasetv2` as the test set; build batch interface (drag folder → single CSV).

T14 — Personalised Indian Recommender

The problem. Recommendation systems are the most-used ML technique in industry. Build a small but real one for an Indian-flavour domain.

What you will build. Streamlit app: query input (text or chips) → ranked card list → "why?" below each item → "save to favourites".

Skills. Sentence embeddings, cosine similarity retrieval, LLM justification generation, no training.

Common base. `sentence-transformers/all-MiniLM-L6-v2` for embeddings. `faiss-cpu` or `chromadb` for retrieval.

Variants

ID	Variant	Tier	Specific dataset
T14.1	"Cook with what I have"	**Tier 1**	`kaggle.com/datasets/kanishk307/6000-ind` (6,000+ recipes with ingredient lists, cuisine, course, diet).
T14.2	Indian-author book recommender	**Tier 1**	Open Library API: `https://openlibrary.org/developers/api`. Filter by Indian authors via the `author/india` subject.
T14.3	Weekend trip recommender	**Tier 1**	GeoNames India: `https://download.geonames.org/export/dump/IN.zip` (free, all populated places). Combine with Wikipedia "List of tourist attractions in [state]".
T14.4	Bollywood / South movie recommender	**Tier 1**	TMDB API (free, register for key): `https://www.themoviedb.org/documentation/api`. Filter by original language (`hi`, `ta`, `te`, `ml`).
T14.5	Engineering college recommender	**Tier 1**	NIRF rankings PDFs (free): `https://www.nirfindia.org/Rankings/2024/EngineeringRanking.html`. Parse PDF → structured CSV.
T14.6	Recipe-by-region recommender	**Tier 1**	`kaggle.com/datasets/nehaprabhavalkar/in` (255 dishes with region, course, diet metadata).
T14.7	Multi-domain personal concierge chatbot	**Tier 2**	Combines 3+ of the above sources behind a single chat interface.

Quick Reference: All 103 Variants by ID

You will enter the ID (e.g. `T7.3`) in the assignment form.

ID	Variant	Tier	Team
T1.1	Tomato disease detector	1	1–3
T1.2	Rice disease detector	1	1–3
T1.3	Potato disease detector	1	1–3
T1.4	Cotton disease detector	1	1–3
T1.5	Sugarcane disease detector	1	1–3
T1.6	Pepper / chili disease detector	1	1–3
T1.7	Multi-crop unified disease app	2	5–7
T1.8	Mobile / PWA crop disease app	2	5–7
T2.1	North Indian curries classifier	1	1–3
T2.2	South Indian breakfast classifier	1	1–3
T2.3	Indian sweets classifier	1	1–3
T2.4	Street food classifier	1	1–3
T2.5	Indian breads classifier	1	1–3
T2.6	Regional thali identifier	1	1–3
T2.7	80-class food + nutrition app	2	5–7
T2.8	Diabetic-friendly meal scorer	2	5–7
T3.1	Devanagari digit recognizer	1	1–3
T3.2	Devanagari character recognizer	1	1–3
T3.3	Tamil character recognizer	1	1–3
T3.4	Telugu vowel recognizer	1	1–3
T3.5	Bengali character recognizer	1	1–3
T3.6	Kannada-MNIST digit recognizer	1	1–3
T3.7	Multi-script router app	2	5–7
T3.8	Printed-text textbook OCR	2	5–7
T5.1	Indian traffic sign classifier	1	1–3
T5.2	Helmet detector	1	1–3
T5.3	Pothole spotter	1	1–3
T5.4	Vehicle counter	1	1–3
T5.5	Lane-line detector (OpenCV)	1	1–3
T5.6	License plate cropper + OCR	2	5–7
T5.7	Road safety dashboard	2	5–7
T5.8	Dashcam highlight extractor	2	5–7
T6.1	5-asana yoga classifier	1	1–3
T6.2	Surya Namaskar step counter	1	1–3
T6.3	Pushup rep counter	1	1–3
T6.4	Squat form checker	1	1–3
T6.5	Plank timer + form check	1	1–3
T6.6	20-asana classifier + form coach	2	5–7
T6.7	Personalised home-yoga app	2	5–7
T6.8	Desk-posture monitor	2	5–7
T7.1	Indian birds (25 species)	1	1–3
T7.2	Indian mammals identifier	1	1–3
T7.3	Snake (venomous vs not) identifier	1	1–3
T7.4	Butterflies identifier	1	1–3
T7.5	Big cats identifier	1	1–3
T7.6	90-animal general identifier	1	1–3
T7.7	100-species fine-tuned identifier	2	5–7
T7.8	Camera-trap auto-tagger	2	5–7
T8.1	Hindi product review sentiment	1	1–3
T8.2	Tamil restaurant review sentiment	1	1–3
T8.3	Telugu movie review sentiment	1	1–3
T8.4	Bengali book review sentiment	1	1–3
T8.5	Hinglish e-commerce sentiment	1	1–3

ID	Variant	Tier	Team
T8.6	Marathi tweet sentiment	1	1–3
T8.7	Multi-language analyzer + dashboard	2	5–7
T8.8	Aspect-based sentiment	2	5–7
T9.1	English news daily briefing	1	1–3
T9.2	Hindi news daily briefing	1	1–3
T9.3	Tech news tracker	1	1–3
T9.4	Sports highlights briefing	1	1–3
T9.5	Stock-market news tracker	1	1–3
T9.6	Translated regional news	1	1–3
T9.7	Bias spectrum visualiser	2	5–7
T9.8	Personalised newsletter generator	2	5–7
T10.1	Passport assistant chatbot	1	1–3
T10.2	Voter ID assistant chatbot	1	1–3
T10.3	RTI filing helper	1	1–3
T10.4	Income tax FAQ chatbot	1	1–3
T10.5	Driving licence & RC chatbot	1	1–3
T10.6	Ayushman Bharat eligibility chatbot	1	1–3
T10.7	Multi-service unified gov assistant	2	5–7
T10.8	IIIT-H rules / hostel / academic bot	2	5–7
T11.1	Hindi digit voice command	1	1–3
T11.2	Tamil basic commands	1	1–3
T11.3	Hindi smart-home commands	1	1–3
T11.4	Bengali navigation commands	1	1–3
T11.5	Telugu emergency phrases	1	1–3
T11.6	Marathi kiosk navigation	1	1–3
T11.7	Wakeword + multi-language commands	2	5–7
T12.1	Top-monuments fine-tuned	1	1–3
T12.2	Forts of Maharashtra	1	1–3
T12.3	Temples of Tamil Nadu	1	1–3
T12.4	Mughal architecture identifier	1	1–3
T12.5	Hampi monuments identifier	1	1–3
T12.6	Stepwells of India	1	1–3
T12.7	100-monument fine-tuned + audio guide	2	5–7
T12.8	Monument scavenger hunt game	2	5–7
T13.1	Resume parser	1	1–3
T13.2	GST invoice parser	1	1–3
T13.3	Marksheet parser	1	1–3
T13.4	Lease agreement parser	1	1–3
T13.5	Prescription parser	1	1–3
T13.6	Bank statement parser	1	1–3
T13.7	HR-tech mini suite	2	5–7
T13.8	Bulk doc-to-database processor	2	5–7
T14.1	Cook-with-what-I-have recommender	1	1–3
T14.2	Indian-author book recommender	1	1–3
T14.3	Weekend trip recommender	1	1–3
T14.4	Bollywood/South movie recommender	1	1–3
T14.5	Engineering college recommender	1	1–3
T14.6	Recipe-by-region recommender	1	1–3
T14.7	Personal concierge chatbot	2	5–7

Want to propose your own project?

The assignment form has an optional “Propose your own” section. Provide: title, problem statement, dataset link, intended app, tier, team size. Faculty/TA approval required before you start work.